

Lab Report 11 – BUOYANCY

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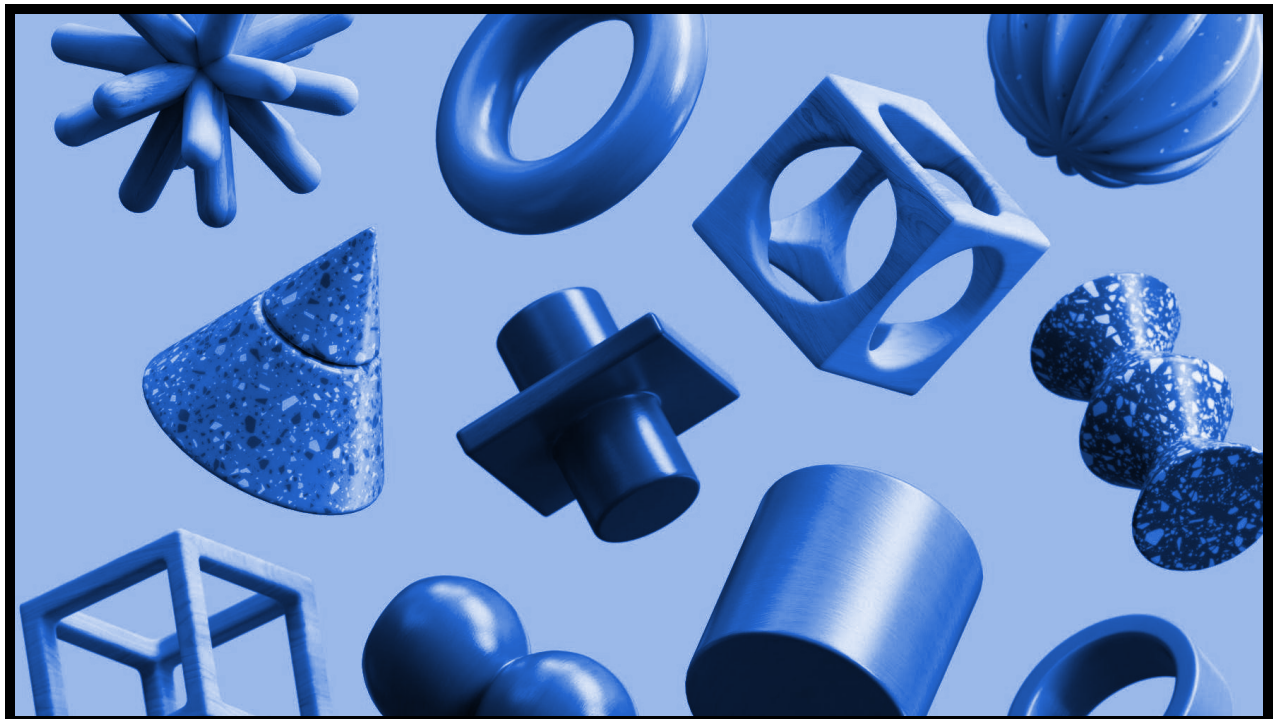
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PHYS 2125

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Objectives

To confirm the theoretical understanding of buoyant force.

Equipments

The list of equipments used in this experiment is:

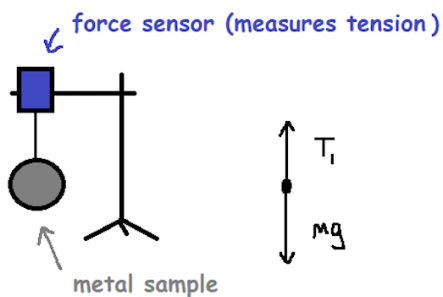
- Lab stand
- Force sensor
- Computer interface
- Digital calipers
- Metal samples
- Beam balance
- String
- Liquid container

Theory and Equations

1. Tension Force in the Air

An object hung from a string is at rest, the tension in the string is equal to the weight of the object.

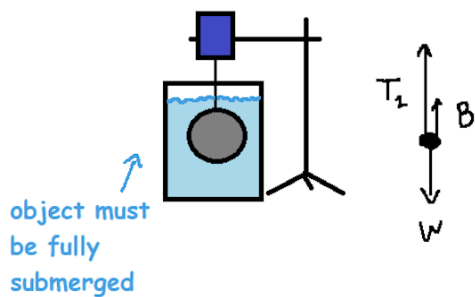
$$T_1 = mg \quad (\text{Eq.1})$$



2. Tension Force in Water

When the object is then submerged in water as shown, the tension in the string now equals the weight minus the Buoyant force:

$$T_2 = mg - B \quad (\text{Eq.2})$$

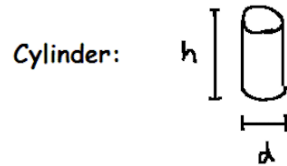


3. Volume Formulas

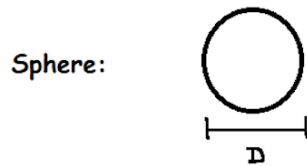
Buoyant force is caused by the volume of fluid displaced.

The volume displaced equals the volume of the object (since it is fully submerged).

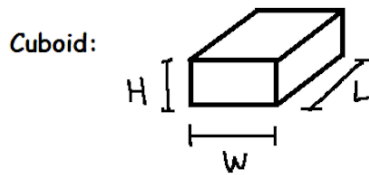
The volume of a sphere, cylinder, or cuboid shape can be easily calculated from its linear dimensions:



$$V_{cylinder} = \frac{\pi}{4} d^2 h \quad (\text{Eq.3a})$$



$$V_{sphere} = \frac{\pi}{6} D^3 \quad (\text{Eq.3b})$$



$$V_{cuboid} = L \cdot W \cdot H \quad (\text{Eq.3c})$$

4. Buoyant Force

Buoyant force is equal to the weight of fluid displaced.

The fluid is water, with a density of 1000 kg/m^3 .

$$B = p_w V_{disp} g \quad (\text{Eq.4})$$

$$p_w = 1000 \text{ kg/m}^3, g = 9.800 \text{ m/s}^2.$$

V_{disp} is calculated from Eq.3a, Eq.3b, or Eq.3c.

Also, we have:

$$T_1 - T_2 = B \quad (\text{Eq.5})$$

5. Percentage Error

The percentage error is calculated based on the experimentally determined measure (exp) and the accepted value (acc). And there are 2 types of accepted values which are theoretical value which is predicted from theory and empirical value which is collected from a reliable source.

$$\text{Percentage Error} = \left| \frac{(\text{exp}) - (\text{acc})}{(\text{acc})} \right| * 100\% \quad \text{units: \%}$$

6. Density

To calculate the density (P) of an object which has a mass (M) and volume (V), we use the following formula:

$$P = \frac{M}{V} \quad \text{units: g/cm}^3$$

Summary of Procedures

First, we prepared a stable surface using the equipment listed above, along with three metal samples: a sphere, a cylinder, and a cuboid. We measured the mass of each sample using a beam balance and recorded their linear dimensions using a rule. Next, we set up the lab stand and force sensor as shown in the above images, ensuring that the force sensor was 0 before attaching any samples.

For each sample, we followed the same procedure: the sample was suspended by a string attached to the force sensor. With the sample at rest in air, we began recording data using the computer interface and allowed several seconds to pass to capture force measurements. We then positioned a container of water beneath the sample and fully submerged it while it remained suspended. Again, we allowed several seconds to pass to record the force data while the sample was submerged. Then, we stopped the recording.

From the collected data, we determined the magnitude value of the average force measured while the sample was suspended in air, which corresponds to T_1 in Equation 1. Similarly, we obtained the magnitude value of the average force while the sample was submerged in water, corresponding to T_2 in Equation 2.

Data and Observations

Measurement					Recorded	
Sample	Mass (kg)	Length (m)			Mass in Air	Mass in Water
		D (m)	h (m)	w (m)	Avg Force - T1 (N)	Avg Force - T2 (N)
Sphere	0.2182	0.03800			2.212	1.929
Cylinder	0.2036	0.02215	0.06349		2.029	1.787
Cuboid	0.06570	0.03174	0.04875	0.01589	0.6440	0.4050

Data Analysis

p_w (kg/m ³)	1000
acc g (m/s ²)	9.800

Sample Calculation:

Sphere:

- Volume = $V_{\text{Sphere}} = \frac{\pi}{6} D^3 = \frac{\pi}{6} (0.03800\text{m})^3 = 2.873 \times 10^{-5} (\text{m}^3)$
- Buoyant Force = $B = p_w V_{\text{disp}} g = (1000\text{kg/m}^3)(2.873 \times 10^{-5} \text{m}^3)(9.800\text{m/s}^2) = 0.2816 (\text{N})$
- $F = mg = (0.2182\text{kg})(9.800\text{m/s}^2) = 2.138 (\text{N})$
- $mg - B = 2.138\text{N} - 0.2816\text{N} = 1.857 (\text{N})$
- $T_1 - T_2 = 2.212\text{N} - 1.929\text{N} = 0.2830 (\text{N})$

Cylinder:

- Volume = $V_{\text{cylinder}} = \frac{\pi}{4} d^2 h = \frac{\pi}{4} (0.02215\text{m})^2 (0.06349\text{m}) = 2.447 \times 10^{-5} (\text{m}^3)$

Cuboid:

- Volume = $V_{\text{cuboid}} = L \cdot W \cdot H = (0.03174\text{m})(0.01589\text{m})(0.04875\text{m}) = 2.459 \times 10^{-5} (\text{m}^3)$

Sample	Calculations				
	Volume (m^3)	Buoyant Force (N)	F = mg (N)	mg-B (N)	T1 - T2 (N)
Sphere	2.873E-5	0.2816	2.138	1.857	0.2830
Cylinder	2.447E-5	0.2398	1.995	1.756	0.2420
Cuboid	2.459E-5	0.2410	0.6439	0.4029	0.2390

Results

Sample Calculation:

Sphere:

- Percentage Error (T_1 vs. mg) = $\left| \frac{2.212 - 2.138}{2.138} \right| * 100\% = 3.444 \%$
- Percentage Error (T_2 vs. (mg-B)) = $\left| \frac{1.929 - 1.857}{1.857} \right| * 100\% = 3.889 \%$
- Percentage Error ($(T_1 - T_2)$ vs. mg) = $\left| \frac{0.2830 - 0.2816}{0.2816} \right| * 100\% = 0.5104 \%$

Sample	Mass in Air	Mass in Water	Calculations				Result		
	Avg Force - T1 (N)	Avg Force - T2 (N)	B (N)	F=mg (N)	mg-B (N)	T1 - T2 (N)	Percentage Error (T1 vs. mg) (%)	Percentage Error (T2 vs. (mg-B)) (%)	Percentage Error ((T1-T2) vs. B) (%)
Sphere	2.212	1.929	0.2816	2.138	1.857	0.2830	3.444	3.889	0.5104
Cylinder	2.029	1.787	0.2398	1.995	1.756	0.2420	1.690	1.793	0.9361
Cuboid	0.6440	0.4050	0.2410	0.6439	0.4029	0.2390	0.02174	0.5194	0.8103

Discussion / Conclusion

The results of this experiment successfully confirmed the theory of Buoyancy Force. Through experimental measurements of the tension in air (T_1) and in water (T_2), the buoyancy force applied to each sample (sphere, cylinder and cuboid) was close to the theoretical calculation with all percentage errors less than 4%. These small differences can be due to human error in measuring the dimensions of each sample, the impurity of the water, or the sensitivity of the sensor.

Furthermore, it appears that the mass value affects the values of the tension in air (T_1), in water (T_2) and the gravitational force ($F=mg$) corresponding to the decreasing order of the sample mass: Sphere mass > Cylinder mass > Cuboid mass. It shows that these forces are directly affected by the object's mass.

While the decreasing order of the Buoyant Force corresponds to the decreasing order of the sample volume: Sphere volume > Cuboid volume > Cylinder volume. This clearly confirms that the Buoyancy depends directly on the volume of the displaced liquid, which is in turn determined by the volume of the samples. This statement is consistent with the Buoyancy theory.

Post-Lab Question

Sample Calculation:

Sphere:

$$\text{Exp Density} = P = \frac{M}{V} = \frac{0.2182 \text{ kg}}{2.873 \times 10^{-5} \text{ m}^3} = 7.595 \times 10^3 \text{ (kg/m}^3\text{)}$$

Sample	Calculations		
	Mass (kg)	Volume (m^3)	Exp Density (kg/m^3)
Sphere	0.2182	2.873E-5	7.595E+03
Cylinder	0.2036	2.447E-5	8.322E+03
Cuboid	0.06570	2.459E-5	2.672E+03

As the accepted value of the density of the Sphere is 7595 kg/m^3 and it has a silver color, we can predict it is likely the Stainless Steel material which has the density range from 7480 kg/m^3 to 7950 kg/m^3 (Engineers Edge, n.d.).

About the Cylinder, it has the density of 8322 kg/m^3 and its color is golden, so it is likely the Brass material which has the density range from $8,440 \text{ kg/m}^3$ to $9,700 \text{ kg/m}^3$ (Engineers Edge, n.d.).

Finally, for the Cuboid, it has the density of 2672 kg/m^3 and its color is also silver, then it is likely the Aluminum material which has the density range from 2600 kg/m^3 to 2830 kg/m^3 (Engineers Edge, n.d.).

References

Engineers Edge. (n.d.). Densities of metals and elements table. Retrieved April 11, 2025, from https://www.engineersedge.com/materials/densities_of_metals_and_elements_table_13976.htm.